Crack Detection

EE397 Problem Specification Report

Group 5:

Patricia Ehigiamusoe

Shakirat Olayode

Md Rayhan Mia

Facilitator: Vahid Fakhari

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Department of Electronic Engineering

Maynooth University

ABSTRACT

Infrastructure encompasses many of the facilities and services we use every day, especially in the public sector. Telecommunication towers, gas pipelines, bridges, train tracks and roads are all examples of essential infrastructure we all use every day. All types of infrastructure will have issues at some point. Cracks are one of these issues. Their occurrence is sometimes due to factors like weather conditions, wear and tear over time and poor-quality engineering. If left to grow cracks can lead to detrimental effects on infrastructure as well as people. Ignoring cracks in a derelict apartment building can result in a multitude of injuries and a potential loss to human life. Identifying cracks early can reduce these adverse effects. Regular monitoring is one possible solution. This is the motivation for this project. We will investigate the area of crack detection with the intention and perspective our creating a device that identifies cracks.

Contents

1. INTRODUCTION

Bridges, roads and buildings are all examples of essential infrastructure we see and use in our everyday lives. We depend on these physical structures to support and improve our existence. Public systems like water, sewage, railways and airports are needed in order for a country to function. [1] Infrastructure dictates the way we travel, transport and work.

Any issues with infrastructure would cause complications and in severe cases could lead to major safety issues or injuries. Recent examples of this include the collapse of a bridge in China in July 2024, and a sinkhole in Thailand in September 2025. The collapse of the bridge in a flood prove area of China was attributed to a steel cable failure and resulted in the deaths of at least twelve people with others reported missing. [2] The sinkhole in Thailand was due to the construction of a metro tunnel and underground stations. [3] While investigations are still ongoing, it has been noted that the collapse was caused by the soil above entering the damaged ceiling of an underground station.

This project aims to research common issues found within infrastructure and develop a practical method to prevent further damage and improve safety by providing early detection. We also intend to create a prototype that is accurate but cost effective. Other possible features will be considered based on research and customer feedback. This will enable us to develop a solid framework to build our project on.

For the purposes of this report, a “crack” will be used to refer to a number of discontinuities within a material. This includes fractures, breaks, faults, etc.

2. LITERATURE SURVEY

In order to improve our understanding of how an infrastructure inspection tool should work, we will be looking at various established solutions. These examples will help us make informed decisions when designing a prototype in the following areas:

- Which sensors would be most appropriate for the device
- Creating a 3D map that shows cracks
- Features to include in the device
    - Displaying information/communicate information to the user
    - Recording and storing data
    - Transforming data into a simulation

2.1 Sensors

While cracks on the surface can be detected by eye, internal cracks are much harder to identify. Both surface and internal cracks should be properly investigated to assess the damage. Non destructive testing (NDT) is a method used to inspect cracks without causing further damage. [4]

Ultrasonic Testing

Ultrasonic testing involves the use of ultrasonic waves to check for gaps under the surface of a material. [4] Sound waves are sent through a material by a transmitter. If no gaps are found, the waves are reflected back and received by a receiver. If a gap is found the waves will be reflected back faster. This kind of testing can be very accurate, but its sensitivity means it may not be suitable to use on a non-smooth surface. The sound waves will be reflected back at different points leading to inaccurate results. However, the ease of use and precision of ultrasonic testing make it a popular option when it comes to crack detection.

LiDAR/Creating a 3D Map

LiDAR (Light Detection and Ranging) is another method that is used to measure distances using laser beams. [5] Similar to ultrasonic testing, LiDAR works by sending and receiving waves, in this case light waves. The time taken for the light to return to the sensor is compared to the constant speed of laser light. This allows the distance travelled to be calculated. A LiDAR point cloud is formed from the repetition of this process. [6]

Point cloud data has 3D mapping capabilities, which makes it suitable for mapping surfaces. While this technique is extremely accurate for measuring distances, it is not great for finer details. A solution to this limitation is to combine LiDAR with other sensors. For example, LiDAR and a depth camera can be used together to provide better mapping results. [7] The depth camera can utilise machine learning algorithms to further improve results along with LiDAR. However, these complex networks require a great deal of computation and therefore is not a simple or cost-effective solution to implement.

2.2 Features

Displaying/Communication Information

There are many different types of crack detector devices available. Most devices typically feature a screen and a probe. [8] The screen may display real time information. Other outputs like sounds and flashing LEDs may be used to communicate things like the device needing to be charged or detecting a crack. Visual and auditory cues often improve a user's experience with a device, instead of relying only on reading information from the screen. For example, if a battery powered device displays a low power message it can be ignored and forgotten about by the user. Whereas it is much more difficult to dismiss a regular beeping sound.

The monitor for a device using ultrasonic testing might display a crack using a graph. The intensity of the crack is indicated on the y-axis and is plotted against time on the x-axis. [9] The initial transmission of the sound wave is first seen in the graph. Then the reflection of the wave when it reaches the end of the test material is seen at the end (back wall). If a crack is detected, it is shown in between the initial probe and the back wall. The amplitude of the signal depends on the depth of the crack.

Recording and Storing Data

Recording and storing real-time data, allows for the data to be utilised more effectively. When a crack is found it would be useful for the device to record certain values. The value at the peak of the crack and the largest width of the crack would be two values of interest. The cracks could then be categorised depending on these values, with categories such as minor, moderate and severe. Of course, such definitions would need to be based on extensive research and data. Additionally, solutions for each class could be suggested by the device. For example, for a small/minor crack perhaps a suitable solution could simply be to fill it in. Recorded data points can also be used after the initial analysis and after the user has left the site. This data could form part of a report on the inspected crack.

Running a Simulation

Any resilient and sustainable infrastructure must be put under rigorous testing before it can be used. Some tests can be carried out quite successfully on components separately, but this can leave room for the overall build to not function as intended once fully constructed. This is where simulations come in. Simulations are used to replicate real-world events without any real-world consequences. [10] This can be used to test materials or components in important but tough to recreate scenarios. For example, individual steel cables used in bridges can be put under tensile test to identify weaknesses, but how can the entire bridge be effectively tested? A simulation of the bridge could be the answer but only if the simulated environment very closely matches real life.

3. CUSTOMER INPUT

We recognise that the target audience of crack detector devices would mostly be professionals. As students we ourselves do not have the experience needed to properly identify what is needed in such a device. Therefore, we conducted a short survey to help us cover this gap in our knowledge. The results of this survey are detailed below.

We assumed many people working in industries like construction, manufacturing, quality control and engineering would be familiar with crack detector tools. The first question in the survey was asked to confirm this. 80% of our respondents said they have heard of crack detector tools.

All participants also said they would trust a device to detect cracks. This attitude shows a level of trust in technology. This could be due to technology becoming more advanced with more accuracy than equivalent manual processes. Technology also usually takes less time to perform actions than a person would.

A graph showing the desired features in a crack detector device is shown below (graph 1). The participants were able to choose multiple options.

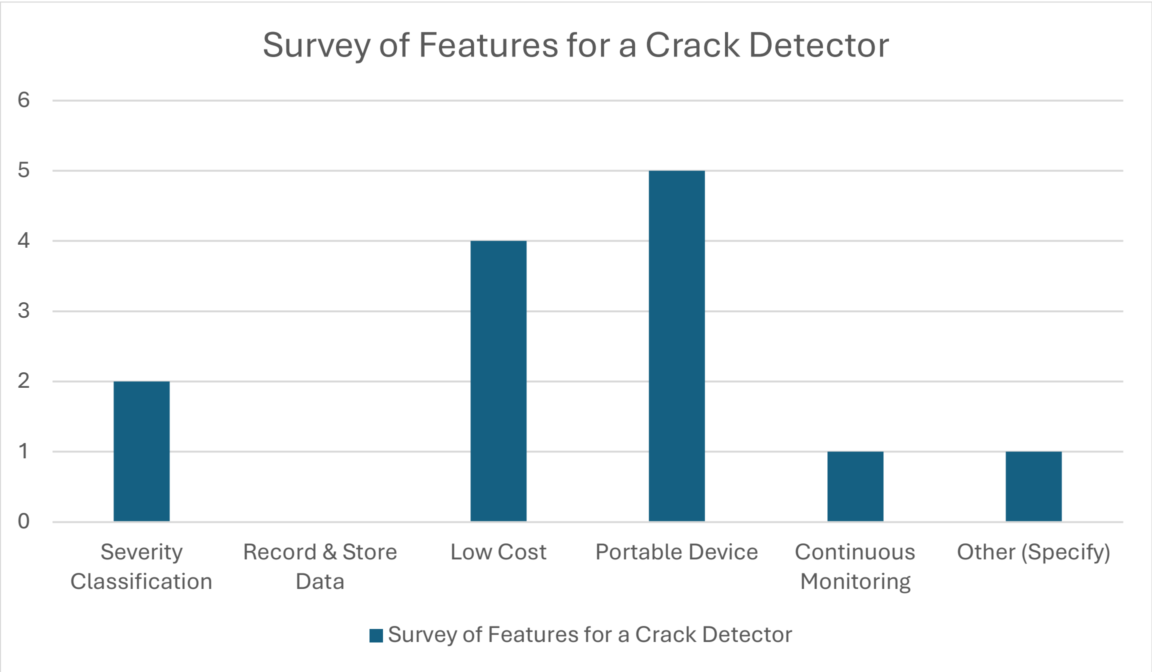

Figure 1. Desired Features in a Crack Detector

The most popular responses were low cost and portable device. 80% of respondents choose low cost and 100% of respondents choose portable device. From research, we found that the average price range of a crack detector device started at around €100 [11] and just went up from there. Top quality detectors can cost up to €7,000. [12]

When asked how likely they would be to use the device and where they would use them, most respondents said they would use it at work. Some also said they would use it at home for home projects.

By conducting the survey, we have identified that there is a market for a cheaper product. There is also the option to make the device smaller, so it is more portable. A cheaper device that doesn't take up much space will also appeal to people involved in home improvement /DIY projects. This opens up a new audience that was not previously considered. We will base our prototype design on the data provided by this survey.

SURVEY 02

A total of 51 participants completed the 2nd survey, representing a diverse group of people from both technical and non-technical backgrounds. This mix ensures that our findings reflect the views of the wider community and not just engineering professionals.

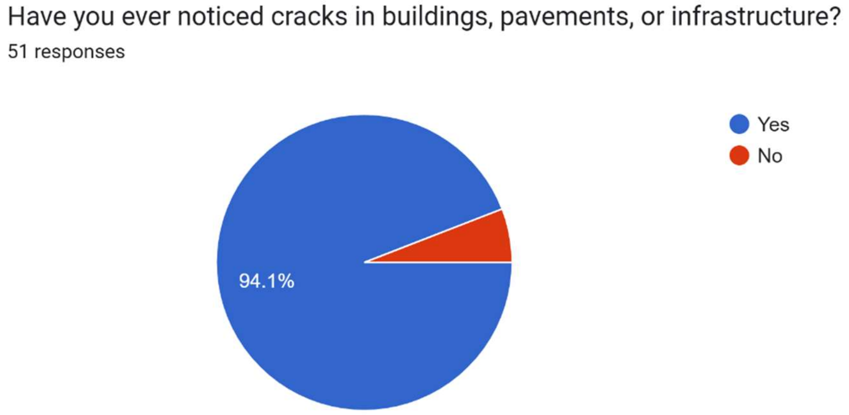

Figure 2. Occupation of survey respondents (n = 51)

Occupation:

The pie chart above reveals that, about 63 % of respondents were students, followed by engineers and architects at 15.7 % the public at, about 11.8 % and a smaller group comprising engineering students and maintenance/facility staff. This shows that although the majority of participants came from a background, we still received input from folks who work hands-on in maintenance and technical roles. That blend of perspective helped our aim to create a system that's both educational and practical.

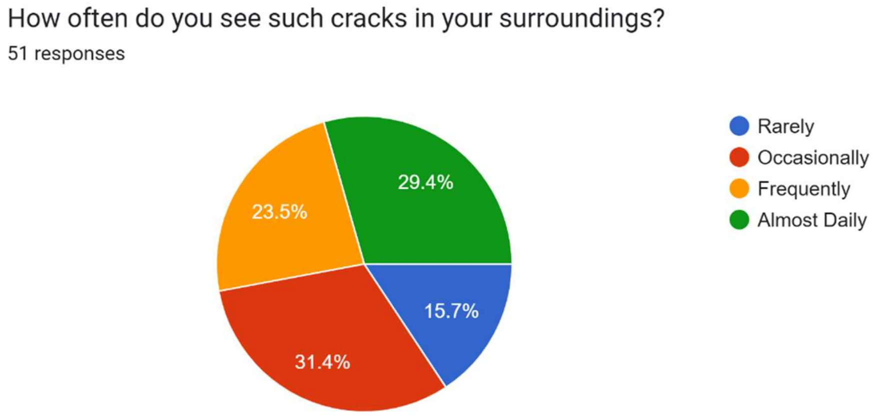

Figure 3. Age distribution of respondents (n = 51)

Age Group:

The second chart shows that the group of respondents—, about 61 %—fall in the age range making it the dominant slice. The next sizeable group, covering a quarter of the sample (25.5 %) spans ages 26-40 while those 41-60 account for just under ten percent (9.8 %). Participants under 18 form a fragment of the total. The age breakdown fits the university setting of our questionnaire. It also reveals that the effort attracted responses from career-oriented professionals and seasoned participants—vital, for assessing usability across a spectrum of ages.

Overall, the survey pulled in a ranging group—students and facility staff, on the user side plus the infrastructure caretakers such as engineers and maintenance teams. That mix lends credibility. Makes the results more relevant, for the sections that follow.

AWARENESS & BEHAVIOUR

To understand how familiar people are with cracks in their surroundings and how they respond to them, we asked participants a series of questions about their awareness and actions.

The questions included:

- Have you ever noticed cracks in buildings, pavements, or infrastructure?
- How often do you see such cracks in your surroundings?
- When you see a structural crack, what do you usually do?
- Do you think cracks can indicate serious safety issues?
- Are you aware of any technology or device that detects cracks automatically?

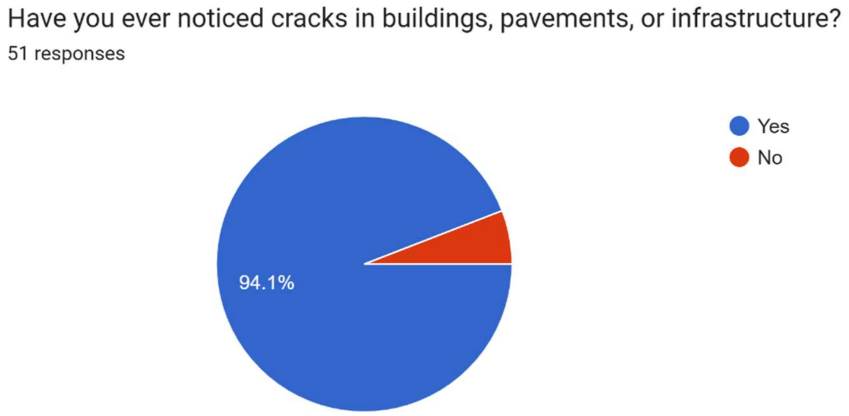

Figure 4. Awareness of cracks in buildings, pavements, or infrastructure

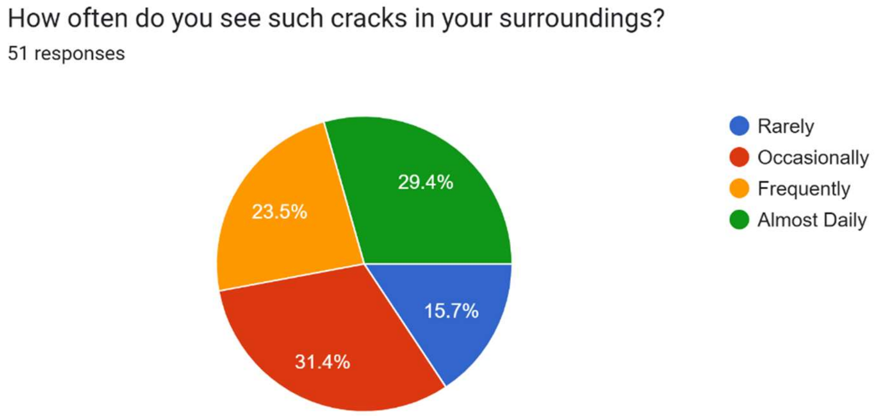

Figure 5. Frequency of observing cracks in surroundings

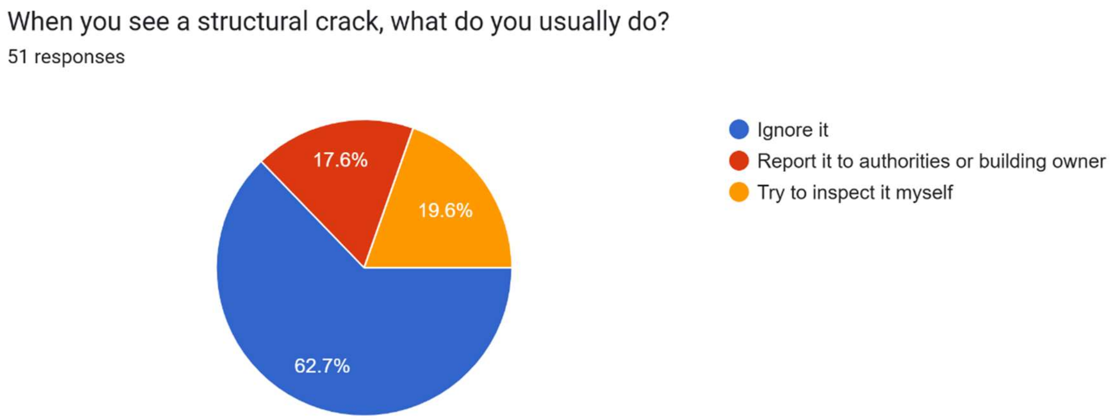

Figure 6. Usual action when noticing a structural crack

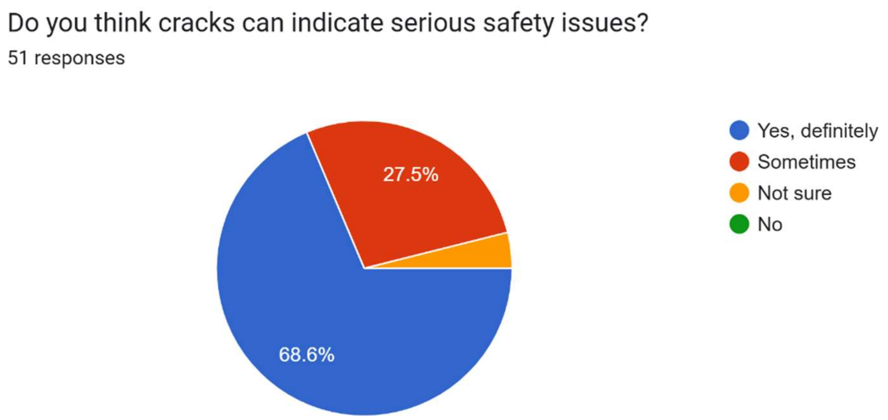

Figure 7. Perception of cracks as potential safety issues

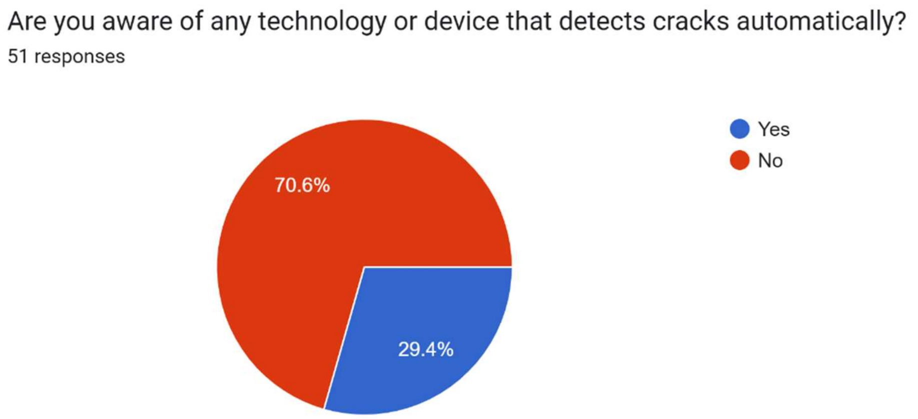

Figure 8. Awareness of existing crack detection technologies

Taking together the findings suggest that a substantial majority of people are already conscious of cracks—virtually everyone has spotted one at some point. Many encounter them on a basis.. The prevailing tendency is to look the way rather than flag the problem even though most respondents acknowledge that cracks can herald serious structural issues. Moreover a notable share of those surveyed admitted they were unaware of any existing detection technology underscoring the opportunity, for a accessible system like SSC-DMS to bridge a significant gap, in both awareness and reporting.

FEATURE PRIORITIES

To identify which features people would find most useful in a crack-detection system, we asked participants what options they would like included.

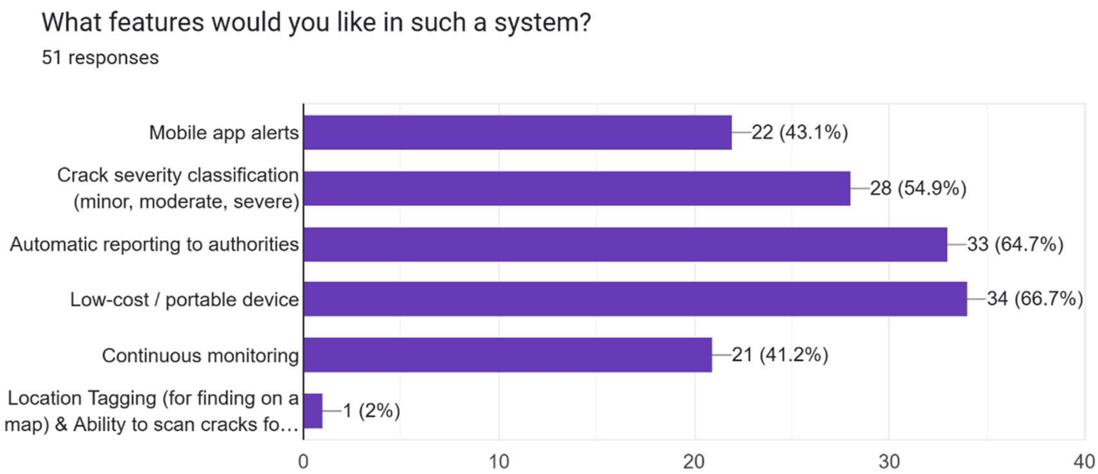

Figure 9. Preferred features in a crack detection system

The survey data point to two features that topped the wish-list: a device—chosen by 66.7 % of respondents—and an automatic reporting link, to authorities favored by 64.7 %. Those numbers underline an appetite for a low-priced tool that can seamlessly flag safety issues. Moreover a little more, than half of the participants (54.9 %) also called for a crack-severity classification hinting that users would value a system of telling whether a crack is merely superficial or potentially serious. Mobile-app alerts (43.1 %) and continuous monitoring (41.2 %) emerged as the cited features while only a modest handful mentioned location-tagging and scanning capabilities. Taken together the responses highlight that users prioritize accessibility, automation and affordability—three core tenets that will shape the SSC-DMS prototype's design.

ADOPTION WILLINGNESS

To evaluate people's trust in technology and their likelihood of using the proposed system, participants were asked about the importance of early detection, their trust in a smart monitoring system, and whether they would use or recommend such a tool.

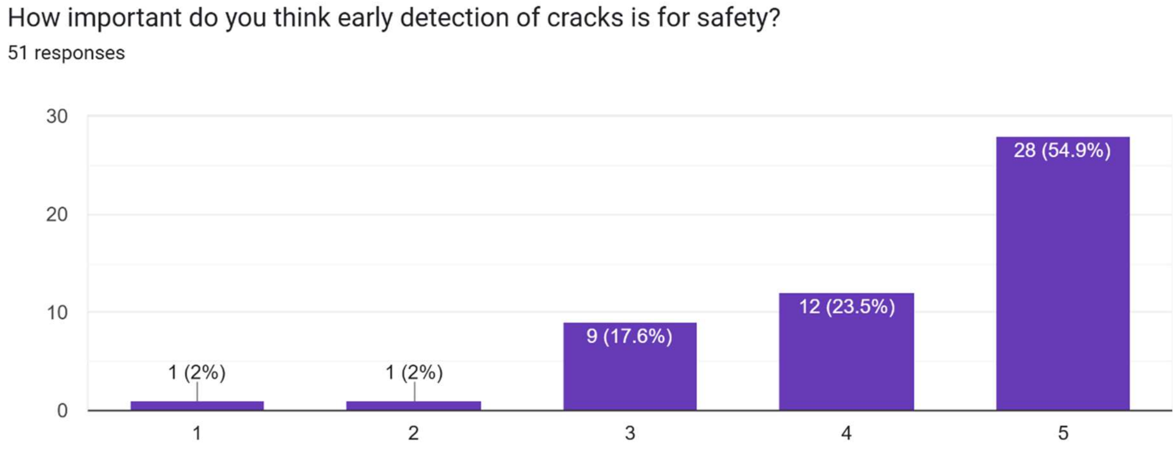

Figure 10. Importance of early crack detection for safety

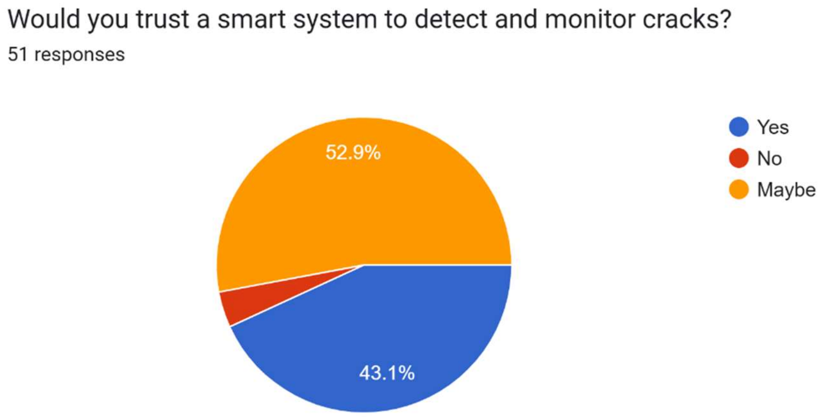

Figure 11. Trust in a smart crack detection and monitoring system

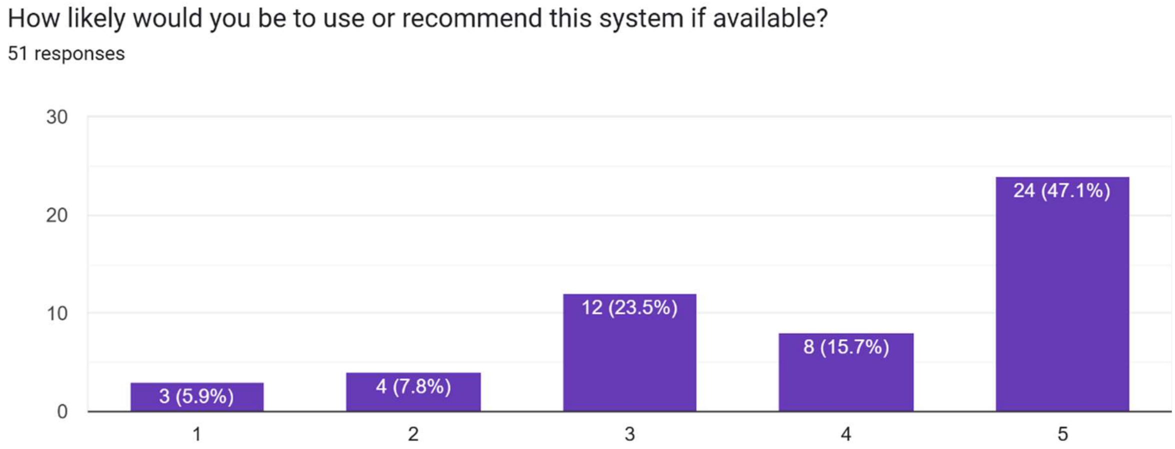

Figure 12. Likelihood of using or recommending the system

In sum the feedback skews decidedly positive. Just, over half of those surveyed—54.9 %—rated crack detection as " important," underscoring a strong safety consciousness. When asked whether a smart detection system would be trusted 43.1 % answered yes. A further 52.9 % responded "maybe " indicating an openness, to new technology if it proves reliable.

Ultimately almost half—47.1%—of those surveyed said they'd be very likely to use or recommend the system once it becomes available. That signals interest and a willingness to embrace affordable accessible options such, as SSC-DMS especially if the system proves reliable and easy to handle.

FEEDBACK & APPLICATION AREA

To understand where the proposed crack detection system would have the greatest impact, participants were asked which types of areas or structures they believe need it the most.

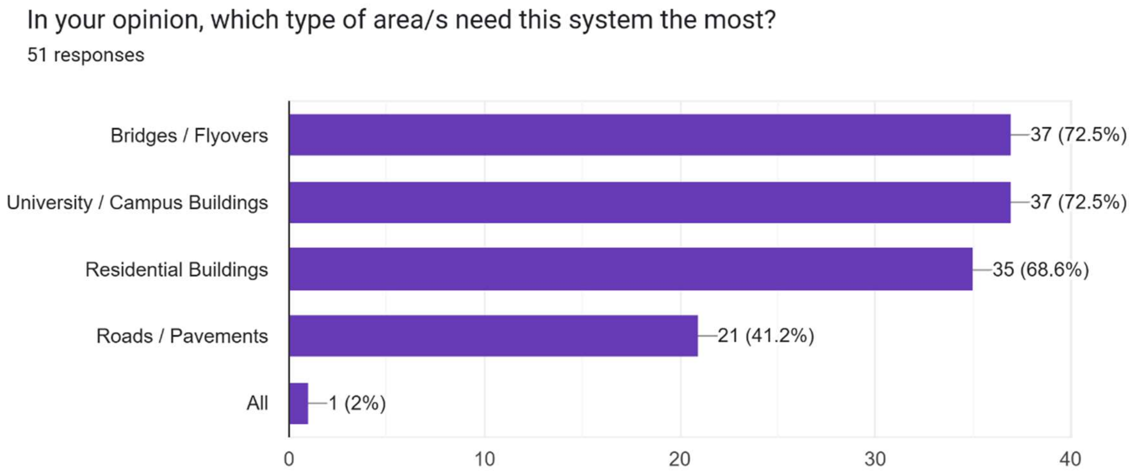

Figure 13. Participant opinions on areas most in need of crack detection systems

The survey shows a consensus: bridges and flyovers (72.5 %) and university or campus buildings (72.5 %) are seen as the places most, in need of the system. That mirrors concerns, about safety and the need to keep used or aging structures well maintained.

68.6 % of respondents chose buildings underscoring that they see crack detection, as more, than a purely engineering problem—it's a concern that reaches into homes and neighborhoods. Roads and pavements attracted 41.2 % of the votes; while deemed urgent they're still regarded as important especially in areas that see heavy public traffic.

All things considered the feedback points to a well-founded perception that SSC-DMS has the flexibility to operate in a variety of arenas—whether it's the backbone of services or the everyday nooks we frequent—and that academic campuses serve as a proving ground, for crafting and vetting the system.

Lastly, we asked the participants if they had any suggestions or comments about our project idea. A few of the suggestions are listed below:

Any suggestions or comments about our project idea?

- "Very interesting. Will be extremely helpful in preventing accidents"
- "Make sure the system have no bug"
- "Focus on ease of use will make this considerably more attractive. If anyone could find, scan and report a crack for example"
- "If its implemented, it will be easier for people to determine cracks. It will be helpful. Looking forward."
- "Stear it to a specific type of infrastructure, e.g. buildings or bridges etc. easier if it's not too broad"

- "I mainly see this as something you would use for eaither under water earthquake, crack detection in bridge or underwater structure that could be in need of this or used in determining the serverity of a buildings damages, so steps can be taken to either tear down for safty reason or find weekness in certain aspects like week pillers that lost its support function due to crack most building are tought out in its design and meterial so taht it doesnt crack or bend under compression or tension. But we can use tilted tower as example why is it tilted could your device monitor or determin serverity or why this happened and how is there a way that your device can alert early on; so the steps may be taken in preventing such an event early on."

4. PROBLEM SPECIFICATION

In the wake of the research and survey the projects ambition is to engineer a Smart Structural Crack Detection & Monitoring System (SSC-DMS) that not identifies but analyses and classifies cracks, across a range of infrastructure. By marrying image scrutiny with sensing the system aims to uncover both conspicuous surface fissures and hidden internal discontinuities that typically evade visual inspection.

The system will employ a smartphone camera as the means of capturing images of surfaces—walls, pavements or concrete structures. Those pictures will then be processed via image-processing and machine-learning techniques to pinpoint the presence of cracks gauge their dimensions and evaluate how severe they are. Complementing the camera sensors such, as vibration detectors will be added to monitor changes, in the material and verify any internal cracks.

All the captured details—crack size, type, location and depth—show up on a interface and get stored digitally for future reference. With that data, in hand trend analysis and maintenance planning become straightforward letting users keep an eye on crack growth over time.

At its core the system pursues three pillars—precision, affordability and broad accessibility. By shaping a prototype that stays inexpensive and easy to carry it opens the door, for not engineers but students maintenance crews and ordinary members of the public to operate it safely and effectively.

In the end the SSC-DMS is intended to function as a preemptive-maintenance gadget catching faults early and thus cutting down the odds of a collapse. What the project aims to deliver is an prototype plus a software model that illustrate how on-the-fly crack detection and perpetual monitoring can raise safety levels and stretch the functional lifespan of community infrastructure.

5. WORKPLAN & GANTT CHART

In order to effectively manage our project, we created a Gantt chart that outlines our workplan. This will help us keep track of tasks and complete the project and reports before the due date. If for any reason we fall behind or end up ahead of our projected schedule, the Gantt chart can be used to ensure we did not miss any tasks.

The Gantt chart below (Figure 1) details our workplan from the beginning of the semester till the project due date (Monday 15th December). From week 3 to week 11, we will have weekly meetings with our facilitator, Vahid Fakhari. From previous projects we know how valuable regular advice and feedback is to a good project. An agenda is created before each meeting and minutes are taken during the meeting. We will not have scheduled lab time to work on the project in week 5 as it's a study week. However, progress can still be made as we develop various prototypes online using TinkerCAD. Towards the end of the project, we will work on the technical report as we finish up the project. This will give us enough time to produce a thorough and well thought out report without the stress of a looming deadline. As a group we are confident we can stick to our guideline and present a successful project within the given timeline.

Figure 14 – Gantt Chart

6. CONCLUSION

This study set out to examine the mounting demand, for practical easy-to-use methods of health monitoring across both private infrastructure. By combing through research and parsing survey responses we discovered that cracks rank among the common and worrisome structural issues reported by the public. Nevertheless many people. Overlook these fissures. Simply lack knowledge of how to address them and few have access, to tools capable of detecting them accurately.

Our deep dive, into the tech there—ultrasonic testing rigs, LiDAR scanners and a host of data-recording setups—served up a double dose of insight: a litany of constraints that hobble current crack-detection methods and a glimpse of the untapped promise they still hold. It also underscored an opening for a kit that fuses smartphone-based visual analysis with sensor input, a hybrid that could span the chasm, between specialist inspection gear and grassroots community-level awareness.

The latest poll of respondents reaffirmed that a sizable slice of the sample grasps the stakes of catching cracks and is inclined to adopt a detection kit provided the price is modest and the interface is intuitive. That finding amplifies the weight of our venture. Spotlights the pressing demand, for fresh breakthroughs, in this niche.

As progress continues the Smart Structural Crack Detection & Monitoring System (SSC-DMS) aims to meld camera-based image processing, ultrasonic sensing and straightforward data-logging into a device. This integration could streamline inspections reinforce safety and support preventive-maintenance initiatives.

In sum our work illustrates that fusing sensing techniques, AI-driven analysis and community participation can deliver a high-impact remedy for a long-standing engineering hurdle. With refinement the SSC-DMS could evolve into a tool, for protecting and prolonging the life of infrastructure across university campuses, local communities and even farther afield.

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8. APPENDIX

Appendix A: Survey Responses

Note: Although all responses are anonymous, each respondent is someone who has experience in construction, engineering, manufacturing, etc. The sample covered various levels of expertise, ranging from DIY home projects to industry professionals.

Survey Response 1

1. Are you aware of any technology or device that detects cracks?
   No, I have never used a device like that.
2. Would you trust a device to detect cracks?
   Yes, I've used devices like stud detectors before, and they work well.
3. Which of these features would you like in such a device?
   a. Crack severity classification (minor, moderate, severe)
   b. Record and store data from session to playback later
   c. Low cost
   d. Portable device
   e. Continuous monitoring
   f. Other (Specify)

   c - portable device, d - low cost
4. How likely would you be to use this device?
   I probably wouldn't use it. I usually just fill cracks at home without giving it too much thought so I wouldn't really be interested in getting one.
5. Where would you use this device?
   I'm not in construction or anything like that by profession so I would probably only use it at home if I had to get one.
6. Any other comments/suggestions?
   How much would a device like this usually cost?
   Responded: The lowest price I've seen is about €100.

Survey Response 2

1. Are you aware of any technology or device that detects cracks?
   Yes, but I have never used one before.
2. Would you trust a device to detect cracks?
   Yes, especially when it's not easy to see, like inside a material.
3. Which of these features would you like in such a device?
   a. Crack severity classification (minor, moderate, severe)
   b. Record and store data from session to playback later
   c. Low cost
   d. Portable device
   e. Continuous monitoring
   f. Other (Specify)

a – severity classification, c – low cost, d – portable, e – continuous monitoring

4. How likely would you be to use this device?
   Yes, it would be useful for work.
5. Where would you use this device?
   I would use it at work my workplace and at home for personal projects.
6. Any other comments/suggestions?
   No

Survey Response 3

1. Are you aware of any technology or device that detects cracks?
   Yes, I've used tech like that.
2. Would you trust a device to detect cracks?
   Yes, technology is always more reliable than people.
3. Which of these features would you like in such a device?
   a. Crack severity classification (minor, moderate, severe)
   b. Record and store data from session to playback later
   c. Low cost
   d. Portable device
   e. Continuous monitoring
   f. Other (Specify)

   d – portable, f – Other: highly accurate and reliable
4. How likely would you be to use this device?
   Very likely, but only if I found it better than the equipment I use now.
5. Where would you use this device?
   I work in quality assurance, so I use tools like this at work.
6. Any other comments/suggestions?
7. ***It's very useful to be able to see exactly where a fault is early in production.***

Survey Response 4

1. Are you aware of any technology or device that detects cracks?
   Yes, I know of a couple tools that can detect cracks.
2. Would you trust a device to detect cracks?
   Yes, manually inspections can be really time consuming.
3. Which of these features would you like in such a device?
   a. Crack severity classification (minor, moderate, severe)
   b. Record and store data from session to playback later
   c. Low cost
   d. Portable device
   e. Continuous monitoring
   f. Other (Specify)

   a – classification, c – low cost, d - portable
4. How likely would you be to use this device?

It's something I would like to have and not need rather than need and not have.

5. Where would you use this device?

   Probably at home and at work.

6. Any other comments/suggestions?

   No.

Survey Response 5

1. Are you aware of any technology or device that detects cracks?

   Yes, but I don't have one.

2. Would you trust a device to detect cracks?

   Yes, as long as it's decently accurate.

3. Which of these features would you like in such a device?
   a. Crack severity classification (minor, moderate, severe)
   b. Record and store data from session to playback later
   c. Low cost
   d. Portable device
   e. Continuous monitoring
   f. Other (Specify)

   c – low cost, d - portable

4. How likely would you be to use this device?

   I would use it if I had one.

   Responded: Why don't you have one?

   I've looked at a few devices, but they were a bit out of my price range.

5. Where would you use this device?

   I would use it at work.

6. Any other comments/suggestions?

   No.